

Task 01 - Hetzner Backup

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Setup

Sources: [hannes](#) and liberal use of AI.

Tools: Virtualbox, WSL.

Method: Scripted backup.

Starting off

We began by looking through the PDFs and the links provided in those PDFs. Then we went on to install the SSH keys which we immediately tested, though they did not work as we had not yet ran chmod on them.

The first step was simply to get access to the root of the server itself, so after restricting the permissions for the ssh keys we successfully accessed the server, from there it was important to understand the structure of the server and what specifically we would need to backup. For this we use commands like:

```
lsblk  
df -h
```

Which can then lead us to which disk to backup, in this case, /dev/sdb. From there we need to figure out what and how we should script in order to backup the server without needing to shut off services. Luckily the heavy lifting has been done by others already and many of the components needed can be found by looking around online, for example on the [hannes](#) page which provides us with the command to read the source disk, compress the files and then back it up to our preferred destination. We changed it somewhat for easier configuration:

```
dd if="$SOURCE_DISK" bs=1M status=progress | pigz -c > "$BACKUP_NAME"
```

So basically starting off we needed to:

1. Get into the server.
2. Find out what and where exactly it is that we need to backup.
3. Create the script to do the backup.

This was our initial script:

```
#!/bin/bash
SOURCE_DISK="/dev/sda"
BACKUP_NAME="/root/htznr-backup-$(date +%Y%m%d-%H%M).img.gz"
echo "=== Starting disk backup ==="
echo "Source: $SOURCE_DISK"
echo "Destination: $BACKUP_NAME"
dd if="$SOURCE_DISK" bs=1M status=progress | pigz -c > "$BACKUP_NAME"
echo "=== Backup complete ==="
ls -lh "$BACKUP_NAME"
```

The final script ended up like this:

```
#!/bin/bash
SOURCE_DISK="/dev/sda"
BACKUP_NAME="/mnt/backup/kalsko-htznr-backup-$(date
+%Y%m%d-%H%M).img.gz"
echo "=== Starting disk backup ==="
echo "Source: $SOURCE_DISK"
echo "Destination: $BACKUP_NAME"
dd if="$SOURCE_DISK" bs=1M status=progress | pigz -c > "$BACKUP_NAME"
echo "=== Backup complete ==="
ls -lh "$BACKUP_NAME"
```

Execution

For the execution we had a simple setup:

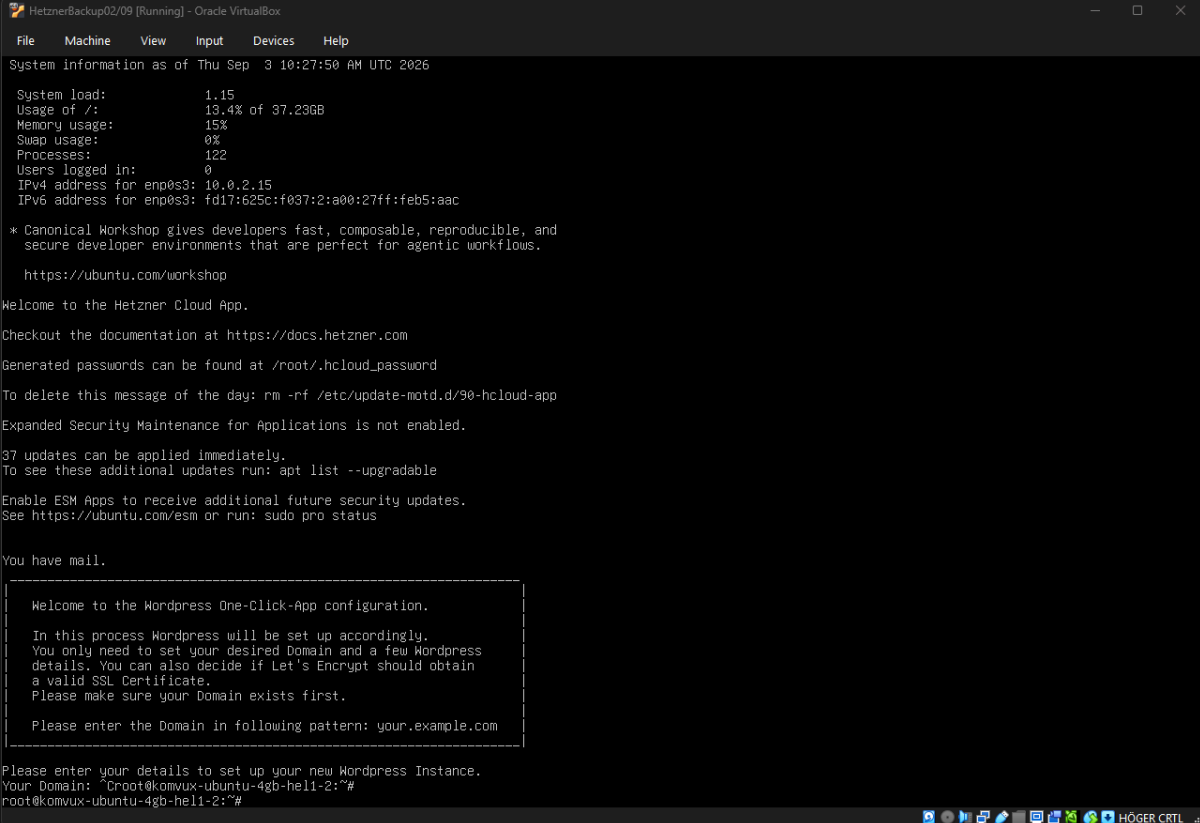
1. SSH into the server.
2. Create the script in the server.
3. Run the script.
4. Exit once the backup has successfully been created.
5. scp the file, gunzip it and convert it to .vdi.
6. Make a new VM using the vdi as its virtual disk, use appropriate settings.
7. Run the VM and hopefully everything boots correctly.

This plan was fine but we did run into some hiccups. First the root folder was full and we had missed the error message initially thinking everything was running fine, we scp'd the file, tried to unzip it and were met with an error message telling us the file ends unexpectedly.

We tested it again with `gzip -t` and got the same result. Going back we removed the previous file and tried to run the script again, this time we saw the error message.

At this point we contacted the teacher who told us the server was full and someone would need to remove the backups already present. While waiting we noticed that the others had been backing up into `/mnt/backup`, realising this we changed the script to personalize the name and send it to `/mnt/backup` instead of `root`.

After sufficient space had been cleared we tried again and this time the backup finished successfully, we `scp`'d it, unzipped it, and converted it successfully and finally had the `.vdi` file we needed. When creating the VM we initially had some trouble as we could not select a `vdi` file, it seems this was because this was the first VM created with virtualbox on this computer so it was not giving us the advanced options right away. After realising this we created another where we could select the `vdi` and successfully booted the VM, but now we were met with a password requirement. As we did not have the credentials we restarted, went into `grep` and then recovery mode and from there we reset the password with `passwd`, rebooted and then logged in successfully to `root`.



```
HetznerBackup02/09 [Running] - Oracle VirtualBox
File  Machine  View  Input  Devices  Help
System information as of Thu Sep  3 10:27:50 AM UTC 2026

System load:          1.15
Usage of /:           13.4% of 37.23GB
Memory usage:         15%
Swap usage:           0%
Processes:            122
Users logged in:      0
IPv4 address for enp0s3: 10.0.2.15
IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:feb5:aac

* Canonical Workshop gives developers fast, composable, reproducible, and
  secure developer environments that are perfect for agentic workflows.

https://ubuntu.com/workshop

Welcome to the Hetzner Cloud App.

Checkout the documentation at https://docs.hetzner.com

Generated passwords can be found at /root/.hcloud_password

To delete this message of the day: rm -rf /etc/update-motd.d/90-hcloud-app

Expanded Security Maintenance for Applications is not enabled.

37 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

You have mail.

-----
Welcome to the Wordpress One-Click-App configuration.

In this process Wordpress will be set up accordingly.
You only need to set your desired Domain and a few Wordpress
details. You can also decide if Let's Encrypt should obtain
a valid SSL Certificate.
Please make sure your Domain exists first.

Please enter the Domain in following pattern: your.example.com
-----

Please enter your details to set up your new Wordpress Instance.
Your Domain: ^Croot@komvux-ubuntu-4gb-hell1-2:~#
root@komvux-ubuntu-4gb-hell1-2:~#
```